UML Software Engineering Project

Aethera — University Edition

AI-Powered Academic Assistant Platform

Student Author: Omar Saeed Mohamed Kamel (AI8 - G3 - 0543)

Program: AI Ambassadors Program (Level 1) — Batch 8

Core Engine: Groq Cloud Llama 3.3 70B Versatile

1. Problem Statement & Vision

University students face heavy information overload from lengthy textbooks, research papers, and complex lectures, requiring immense time to summarize notes, practice exams, and format citations.

🚀 The Vision

Deliver a unified, zero-latency academic assistant that automates summarization, exam generation, mind mapping, and flashcards while offering live Notion synchronization.

⚡ Key Differentiator

Leverages Groq API Llama 3.3 70B for sub-second streaming inference, coupled with local document parsing (PDF/DOCX) for complete student privacy.

2. Functional Specifications

📚 Academic Tools (17+)

Summarizer, Exam Generator, Concept Breakdown, Study Planner, Essay Assistant, Citation Builder, Smart Translator, and Cornell Notes.

💡 Interactive Learning

Flashcard Generator (Anki export), SVG Mind Map Visualizer, Quick Quiz Engine with instant feedback, and AI Chat Assistant.

📊 Productivity Suite

GPA Calculator (Semester/Cumulative), Pomodoro Timer, History Manager, and Notion Workspace Live Sync.

3. Architecture & Tech Stack

Client Layer (SPA)

Vanilla JavaScript & HTML5 Glassmorphic UI with PDF.js & Docx local text extractors.

Server Layer

Node.js & Express REST API Server managing endpoints, security headers, and AI proxying.

Database Layer

SQLite local database (17 structured schemas) & Supabase PostgreSQL with `pgvector` support.

4. UML Analysis & Design Deliverables

The project delivers the complete set of 8 standardized UML diagrams:

1. Context Diagram

Defines system boundary and external entities (Student, Groq API, Notion API, Database).

2. Use Case Diagram

Primary actors, 12 core use cases, and `<<include>>` / `<<extend>>` relationships.

3. Activity Diagram

End-to-end procedural workflow from input parsing to AI stream rendering and storage.

4. Sequence Diagram

Detailed sequence of messages between Student UI, Controller, Groq Engine, and Notion API.

5. Behavioral, Data & Object Modeling

5. State Machine Diagram

State transitions (Idle, Inputting, Processing, Rendering, History Saved, Error).

6. Data Flow Diagram (DFD)

Level 0 Context and Level 1 Detailed DFD showing data stores and transformation processes.

7. Entity Relationship Diagram

17 relational database tables with Primary Keys, Foreign Keys, and Cardinality constraints.

8. Class Diagram

Object-Oriented design detailing class attributes, methods, visibility (+/-), and associations.

6. Front-End Prototype & Conclusion

The front-end prototype is fully functional and live at mindsparkproo.netlify.app.

✅ Engineering Objectives Achieved

100% trace between design UML models and prototype implementation code.

📄 Complete Final Deliverables

Formal Report (PDF), Presentation Slide Deck (PDF), and Working SPA Prototype.